



Panama's oldest marine mammal

A researcher set out to find plant fossils accidentally stumbled upon a 20-million-year-old tusked sea cow fossil. <https://digit.in/mar19-03>



Mountains taller than Himalayas?

Seismologists have discovered a mountain taller than Himalayas 660 km underground <https://digit.in/mar19-02>



Credit: Ryan Sault

STEVE

THE AURORA THAT ISN'T

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The Earth is surrounded by an envelope of gas, and the density of the gas reduces in a gradient into outer space. The atmosphere is massive, and the uppermost layers reach up to an altitude of 190,000 kilometers, or about half way to the Moon. Many natural phenomena occur within the atmosphere, and scientists are only just learning about and studying some of them recently. The aurora borealis and the aurora australis that appear in the arctic and antarctic regions are among the most spectacular natural

light shows on Earth. They appear as winding curtains or diffuse glowing lights in many colours, and are caused by electrons and protons from the solar wind striking the upper atmosphere, specifically the thermosphere and the exosphere. Considering how awe inspiring the aurorae are, there are people who dedicate themselves to chasing them and documenting them. Since the 1950s, scientists have been using the data from these amateur aurora chasers to study and understand the phenomena better. With the availability of high resolution digital cameras, smartphones, as well as satellites, the collaboration between scientists and citizen scientists has grown much more productive in the recent years. For example, in 1997,

scientists would consider themselves lucky if they could correlate one photograph from the ground to a satellite observation, in a single night. Now, they can correlate upwards of a 100 photographs every night.

One group of enthusiasts, known as the Alberta Aurora Chasers, brought a mysterious phenomenon to the attention of space physicists in 2016. This was a purplish arc in the sky, that was not a curtain or a diffuse glow like the other aurorae. The light would appear at lower latitudes than where the aurorae typically appeared. In the animated movie, *Over The Hedge*, a bunch of frightened creatures come across a previously unknown object, a hedge, and begin to call it "Steve", to make it less scary. The enthusiasts started referring to the new phenomenon as Steve. The purplish streak had been photographed before, for decades, and was known as a "proton arc". This is not an accurate description, as specialised equipment is necessary to photograph protons.

After being alerted by the group, scientists began to study Steve in earnest. While aurorae can persist in the sky for hours, Steve is a more ephemeral phenomenon, and is visible for a very short time. It can appear for longer than 20 minutes, but less than an hour. Sightings of Steve have been reported from the Northern US States, Canada, UK and New Zealand. The original name of Steve was adopted by the scientific community, which gave the phenomenon a suitable backronym, dubbing it the Strong Thermal Emission Velocity Enhancement. The naming of Steve was similar to that of other unrelated phenomena in the upper atmosphere, called transient luminous events. These are the sprites (Stratospheric/Mesospheric Perturbations Resulting from Intense Thunderstorm Electrification) and trolls (Transient Red Optical Luminous Lineament), which are also backronyms for mysterious phenomena. While the aurora chasers went out to document sightings of Steve on the ground, the scientists turned